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NPS/001/002 – Technical Specification for Helical Products

1. Purpose

The purpose of this document is to detail the technical requirements for helical products used on the CE Electric distribution networks.

This document supersedes the following documents, all copies of which should be destroyed.

Reference	Version	Title
DSS/004/013	1.0	Specification for Preformed Helical Fittings
MS-O-17	1.0	Technical Specification for Helical Products
NPS/001/002	1.0	Technical Specification for Helical Products

2. Scope

This specification applies to all helical products for use on the CE Electric UK overhead line distribution network and details their range and size where applicable. The majority of items are detailed in national, international and industry standards that are listed within the technical description for each product type.

3. Technical Requirements

3.1 General Requirements

Helical products shall be manufactured and tested in accordance with ENA TS 43-15, ENA TS 43-91 & ENA TS 43-92. Schedule B provides details of the sizes and types of helical products required under this specification.

Helical products shall be factory formed from multiple steel wires, which are generally either aluminium or copper coated for use with bare conductors and normally galvanised for use with PVC covered conductors

Grit shall be applied to the gripping sections of the ties during manufacture to maximise their holding strength.

Where HV top and side ties are to be used with bare conductors, elastomeric pads shall be provided with the ties that fit between the conductors and the insulators to prevent abrasion.

With the exception of LV concentric cable terminations, all other LV tension terminations included in this specification are of the failure containment type. With these fittings, pads are not normally required.

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3.2 Materials

3.2.1 General Requirements

Materials used in the manufacture of conductor fittings shall:

- Withstand the mechanical loads relevant to the installation – service – maintenance conditions, service temperatures and environmental effects.
- Be compatible with the conductor material, or capable of being used with an intermediate material such that there can be no deleterious effect on the conductor or fitting resulting from their association.
- Not adversely affected by any coating applied for corrosion protection.

3.2.2 Materials and Manufacturing

3.2.2.1 Steel Wire

The steel wire to be used for helical fittings shall be manufactured to BS EN 10270 of a composition complying with that specified for type NS. The wire shall be hot dipped galvanised to Galvanising Class 'A' of BS EN 10244. The minimum tensile strength requirement is 1400 N-mm^{-2} .

3.2.2.2 Copper Covered Steel Wire

The copper covered steel wire to be used for helical fittings shall be manufactured to BS 4087. The composition of wires with diameters smaller than 2.59 shall be a minimum of Grade 1 with 40% conductivity. The composition of wires for diameters of 2.59 and above shall be a minimum of Grade 2 with 30% conductivity.

3.2.2.3 Aluminium Covered Steel Wire

The aluminium covered steel wire to be used for helical fittings shall be manufactured to ASTM B415. It shall be composed of a steel core with a substantially uniform and continuous aluminium covering thoroughly bonded to it. The radial thickness of the aluminium coating shall be not less than 5% of the wire diameter at any point. The minimum conductivity of the wire shall be 20.3%

3.2.3 Conductor Pad (for use with ties)

The pad, normally of a split tube design having a nominal wall thickness of 3mm, shall have an inside diameter that will allow it to fit snugly around the diameter of conductor being used. A minimum gap of 1mm shall be left between the closing edges of the pad before application of the fitting. The pad shall meet all of the requirements detailed in ENA TS 43-15 Appendix E, Clause 3.

3.2.4 Grit

To maximise the gripping strength of a fitting, an adhesive compound retaining a chemically inert grit shall be applied during manufacture to the inner surface of the legs where they make contact with the conductor. The loop or crown area of the fitting shall be kept free from the grit.

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3.2.5 Formation of wires

Multiple steel wires coated or galvanised as appropriate, shall be factory formed into a central crown or loop with two legs. For bare conductors, the direction of the helix shall be the same as the direction of lay of the outer strands of conductor to which it is to be applied. For PVC covered conductors, the direction of the helix shall be right hand.

The surfaces and ends of the wires, apart from where grit is applied, shall be free from sharp edges to avoid damage to the conductor or insulator during application and in service.

3.2.6 Protection against corrosion

All materials used in the construction of overhead line fittings shall be inherently resistant to atmospheric corrosion that could affect their performance.

3.3 Types of Fittings

3.3.1 High Voltage (HV) Conductor Tension Fittings

Conductor Dead-ends and Distribution Grip Dead-ends

Conductor Dead-ends and Distribution Grip Dead-ends shall fully meet the requirements detailed in ENA TS 43-92 and BS EN 61284.

The fittings shall be manufactured from material compatible with the conductor to which they are being applied. This will normally be either aluminium or copper clad steel. The dead-ends shall be designed to achieve 100% of the rated breaking strength of the conductor.

The lay direction of the conductor dead-ends shall be the same as that of the outer layer of the conductor to which it is applied.

Armour Splice

Armour Splices shall fully meet the requirements detailed in ENA TS 43-92 and BS EN 61284.

These items shall be designed to repair severely damaged conductors, to give protection and to provide a jointing capability. Armour Splices are normally used to repair conductor damage which has occurred under suspension clamps or pin insulator binds. They shall provide full electrical conductivity to the conductors. The Armour Splice shall also restore full mechanical strength to the conductors when the damage is located under the centre of the repair fitting. On ACSR fittings, the strength shall be restored to all of the aluminium strands but not the core.

Armour splices shall be manufactured from a material that is compatible with the conductor to which they are applied.

The lay direction of the armour splice must be the same as that of the outer layer of the conductor to which it is applied.

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3.3.2 HV Insulator Ties

Groove Formed Side Ties

Groove formed Side ties shall fully meet the requirements detailed in ENA TS 43-15.

Groove formed side ties are intended to secure bare or insulated conductors in the side groove of vertically and horizontally mounted pin type insulators at intermediate angle positions up to the maximum angle of 60° as detailed in ENA TS 43-40.

All fittings shall be provided with neoprene pads to eliminate conductor abrasion at the conductor/insulator interface.

These fittings shall be designed for installation on insulators designed to ENATS 43-93.

HV Top Ties / Distribution Ties

Distribution ties shall be manufactured and tested in accordance with ENA TS 43-15. They shall be designed to secure bare or insulated conductors in the top grooves of vertically mounted pin type 11kV, 20kV or 33kV insulators at intermediate or intermediate angle positions. On vertically mounted insulators, distribution ties shall be suitable for line deviations up to 10 degrees.

All fittings shall be provided with neoprene pads to eliminate conductor abrasion at the conductor/insulator interface.

These fittings shall be designed for installation on insulators designed to ENA TS 43-93.

HV Double Top Tie

Double distribution ties shall be manufactured and tested in accordance with ENA TS 43-15. They shall be designed to secure bare or insulated conductors in the top grooves of vertically mounted pin type 11kV, 20kV or 33kV insulators at intermediate or intermediate angle positions on lines designed to ENA TS 43-40. This fitting shall be constructed from two elements and be designed to provide a higher degree of security especially in locations subjected to vibration or severe weather conditions.

All fittings shall be provided with neoprene pads to eliminate conductor abrasion at the conductor/insulator interface.

These fittings shall be designed for installation on insulators designed to ENA TS 43-93.

Twin-Grip Uplift Ties

Twin Grip Uplift ties shall be manufactured and tested in accordance with ENA TS 43-15. They shall be designed to secure bare conductors in the top grooves of vertically mounted pin type 11kV, 20kV and 33kV insulators at intermediate or intermediate angle positions.

These fittings shall operate like two conductor dead ends, providing additional security to the attachment point when the structure is subjected to a limited uplift situation. Typically these fitting shall be capable of a 10 degree uplift capacity.

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All fittings shall be provided with neoprene pads to eliminate conductor abrasion at the conductor/insulator interface.

These fittings shall be designed for installation on insulators designed to ENA TS 43-93.

3.3.3 Low Voltage Conductor Tension Fittings

Limited Tension Concentric Dead-ends

Limited tension concentric dead-ends shall be designed and tested in accordance with ENA TS 43-92. The fittings shall be designed specifically to terminate low voltage concentric service cables designed in accordance with BS 7870 - 3.10. These shortened fittings shall have a limited tension capability and shall normally be manufactured from galvanised steel.

Service Grip Dead-Ends

Service Grip dead ends shall be designed and tested in accordance with ENA TS 43-92. The fittings shall be designed as limited tension dead ends specifically for the termination of low-tension service spans. It shall be suitable for the termination of 16mm² PVC covered conductor designed in accordance with BS 7884 Table 3.

Low Voltage Dead-End

Low voltage dead-ends shall be designed and tested in accordance with ENA TS 43-92 and be suitable for the termination of single bare or insulated conductors on reel type insulators in accordance with ENA TS 43-93 Fig. 9.1.

3.3.4 LV Insulator Fittings

LV Intermediate/Angle Ties

Shall be designed and tested in accordance with ENA TS 43-15 and be suitable for securing conductors in the grooves of reel type LV insulators at intermediate/angle positions up to the maximum angles given in ENA TS 43-30 for the line design.

3.3.5 Stay Fittings

Guy Grip Dead End

Guy Grip dead ends shall be suitable for the termination of 7/4.00mm grade 1150 galvanised steel stay wire as detailed in ENA TS 43-91 Table 1. All fittings shall provide a MFL of 101kN as detailed in ENA TS 43-91 Table 3.

Insulator Link Assembly

Shall be suitable for joining two porcelain stay insulators in series on unearthed overhead line structures as detailed in ENA TS 43-91 clause 1.4.4.3
All fittings shall provide a MFL of 101kN as detailed in ENA TS 43-91 Table 3.

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Pole Top Make-off

Pole Top Make-Off's are primarily intended for pole top termination of 7/4.00mm Grade 1150 stay strand when an insulator is being used. The fitting shall be manufactured with three strands of high tensile and one strand of medium tensile galvanised steel wire.

When applied at the pole top the completed tail end shall form a stay strand equivalent to the 7/4.00mm stay wire.

The medium tensile quality strand shall be suitable for bonding the helical fitting to the pole top steelwork.

All fittings shall provide a MFL of 101kN as detailed in ENA TS 43-91 Table 3.

Stay Wire Line Splice

Stay Wire Line splices shall be suitable for joining sections of 7/4.00mm grade 1150 galvanised steel stay strand together. The jointed strand shall be suitable for 100% of the original stay strand MFL.

Splices shall be made of material compatible with the galvanised stay strand.

Stay Extension

The stay extension - 1200mm length of 7/4.00mm grade 1150 galvanised steel stay strand.

Stay Make-Off / Pole Top Dead End

This fitting shall be suitable for application around a range of pole diameters. Once installed it shall be suitable for terminating a 7/4.00mm grade 1150 galvanised stay wire to the pole. All fittings shall provide a MFL of 101kN as detailed in ENA TS 43-91 Table 3.

3.3.6 HV & LV Flight Diverters

HV Flight Diverters

HV flight diverters shall be suitable for application onto bare or covered HV lines that have been built in accordance with ENA TS 43-40. The fittings shall ideally be constructed from two separate items.

- A) Orange high visibility ball that is UV stable and suitable for attachment on to a helical conductor fitting.
- B) A helical conductor fitting designed to be attached to the overhead conductors

LV Bird Flight Diverter

LV bird flight diverters shall be suitable for application to LV open wire networks designed in accordance with ENA TS 43-30. The fittings shall be secured to the conductors using a helix rather than fixed point contacts.

The diverters shall be manufactured from an insulating material and be suitable for a range of conductor sizes.

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3.3.7 Other Fittings

Limited Contact Spacer

LV Limited contact spacers shall be designed to be single piece helical fittings that have been designed to prevent damage to conductors arising from conductor clashing. They shall be manufactured from a rigid insulating material. It shall be manufactured using a concave design profile to avoid crossing of conductors.

Semi-conducting Spiral Tape

Semi-conducting spiral tape shall be supplied for use with helical fittings applied to PVC covered conductors. It shall have the value of resistivity, and be applied to the dimensions, as specified in Post Office Memorandum A231 (b) (issued under Engineering Recommendation PO2).

Shunt Splice

The splice shunt shall be designed to restore electrical and mechanical strengths of existing joints that have been found to develop abnormally high resistance. The rods shall perform this function by shunting current around the installed joint and provide additional heat radiation surface.

Spiral Wrap Vibration Damper

Manufacturers shall supply Spiral wrap vibration dampers that are suitable for application onto HV XLPE Covered conductor lines. The dampers shall be suitable for application to the conductor without the need to remove any XLPE insulation. They shall be manufactured from UV stable, high impact insulating material.

3.4 Identification and Marking

Marking shall ensure a system of traceability for each of the component parts of the fittings.

Conductor fittings shall be identified by the manufacturers identification mark and fitting reference. This marking shall also be applied to any component of a fitting where the component is separate from the fitting when dispatched by the manufacturer.

A durable label shall be securely attached to each fitting, containing the following information:

- (i) Fitting type.
- (ii) Size, stranding and type of conductor.
- (iii) Insulator type (if applicable).
- (iv) Fitting identification number.
- (v) CE Electric UK part number.

The labels attached to all LV fittings shall state in bold letters: 'ONLY TO BE USED ON LV LINES'.

Helical fittings shall be colour coded as detailed in ENA TS 43-15 Appendix D, Clause 4.

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3.5 Performance Requirements

3.5.1 HV ties

Resiliency

Ties shall permit controlled movement of the conductors over the insulators under unbalanced load conditions resulting from impact loads on the conductors or support, conductor oscillation, galloping, ice unloading or in the extreme case, broken conductors.

Flashover and withstand voltages

The use of ties on HV insulators shall not reduce the flashover and withstand voltage capabilities of the insulators as detailed in ENA TS 43-93.

Radio interference voltage (RIV)

The design of the ties for use on HV insulators shall provide minimum RIV levels for completed insulator/conductor tie assemblies.

3.5.2 HV Terminations

These shall meet the requirements of ENA TS 43-92 and BS EN 61284

Factory formed helical tension terminations shall be capable of being removed and reapplied to the conductor twice after the original application for the purpose of adjustment during construction procedures.

3.5.3 LV Intermediate/Angle Ties

No movement of the conductors through the ties is permitted for out of balance tensions less than 200kgf.

Under broken wire conditions, controlled movement of bare or PVC covered conductors through the ties is permitted, provided that the out of balance tensions exceed 200kgf.

3.5.4 LV Tension Terminations

No movement of the conductors through the terminations is permitted, if tensions up to twice the maximum design working tension are applied.

Under impact or ice loading conditions, controlled movement of bare or PVC covered conductors through the terminations is permitted, provided that the resultant tensions exceed twice the maximum working tension.

Once controlled movement has occurred, the terminations must not permit further movement of the conductors if tensions up to the maximum erection tensions are reapplied.

3.5.5 LV Concentric Cable Tension Terminations

No movement of the cables through the terminations is permitted, if tensions up to the maximum working tension of 132kgf are applied.

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3.5.6 Specific requirements for Armour Rods

Armour rods shall be of the factory formed helical type and comprise of sets of individual rods compatible with the conductor and shall have right hand lay.

The rods shall completely enclose the conductor except for small gaps required to ensure that the rods make good contact with the conductor over its entire length.

3.6 Type and Sample Testing

Fittings shall be tested in accordance with the requirements of: -

ENA TS 43-15 - Appendix E
ENA TS 43-91 - Clause 8.1.1
ENA TS 43-92 - Clause 9.2

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-15 Issue 3: 2004	Insulator Binds & Equivalent Helical Fittings for Overhead Lines
ENA TS 43-91 Issue 4: 1984	Stay strands and stay fittings for overhead lines.
ENA TS 43-92 Issue 2: 1985	Conductor fittings for overhead lines.
ENA TS 43-93 Issue 4: 2004	Line insulators.
BS EN 1461: 1999	Hot dipped galvanised coatings on fabricated iron or steel articles – specifications and test methods
BS EN 61284: 1998	Overhead Lines - Requirements and tests for fittings.
BS EN 7870 3.10: 2001	PVC insulated concentric service cables with stranded copper or solid aluminium phase conductors and copper concentric conductors.
BS EN 10244 Issue 2: 2001	Specification for testing zinc coatings on steel wire and for quality requirements.
BS 3288 Part 2: 1990	Specification for insulator and conductor fittings for overhead power lines.
ER L38/1	ENA Engineering Recommendation for - Protective greasing of PVC covered copper and bare PVC covered aluminium based overhead line conductors.

4.2 Internal documentation

Reference	Title

5. Definitions

Term	Definition
MFL	Minimum Failing Load

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Steve Salkeld	Standards Engineer	Steve Salkeld	18/12/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Geoff Bolden	Purchasing Operations Manager	Geoff Bolden	20/12/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Ged Hammel	Overhead Standards Manager	Ged Hammel	20/12/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	DBD Administrator	Sean Johnson	11/01/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Chris Holdsworth	Standards and Assessments Manager	Chris Holdsworth	24/12/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Carol Morgan	Head of Procurement	Carol Morgan	27/12/07

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Appendix A – Self Certification Conformance Declaration

Helical Ties covered by ENA TS 43-15 & 43-92 shall comply with the latest issues of the relevant international and British Standards.

ENA TS 43-15 & 43-92 are intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENA TS 43-15, BS EN 61284 & 43-92 and the clauses of the aforementioned Standards relevant to helical Ties for use on current United Kingdom designs of wood pole overhead lines.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENA TS' as appropriate

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Helical Fittings for Insulators

Table 1

ENA TS 43-15			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
4.3	Materials & Manufacture		
Section 4.3.1	Steel Wire manufactured to BS 10270, composition type NS & Hot Dip Galvanised to Class 'A' of BS EN 10244		
Section 4.3.2	Copper Covered Steel Wire manufactured to BS4087		
Section 4.3.3	Aluminium Covered Steel Wire manufactured to ASTM B415		
Section 4.3.4	Copper Alloy Wire manufactured to BS EN 12166		
Section 4.3.5	Pad (For Use with Ties) - See also the requirements of and Appendix A of ENA TS 43-15		
Section 4.3.6	Grit		
Section 4.3.7	Formation of Wires		
Section 4.3.8	Markings		
Section 4.3.9	Semi Conducting Tape		
Section 4.3.10	Grease in accordance with L38		

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Table 2

• ENA TS 43–15			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
Section 4.4	Performance Requirements		
Section 4.4.1	HV Ties		
Section 4.4.4.1	Resiliency		
Section 4.4.1.2	Flashover and withstand voltages		
Section 4.4.1.3	Radio Interference Voltage		
Section 4.4.3	LV Intermediate/angle ties		
Section 4.4.4	LV Tension terminations		
Section 4.4.5	LV Concentric cable tension terminations		

Table 3

• ENA TS 43–15			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
D1	Appendix D - Helical Fittings - Application and Colour Marking		
D2	Fittings for HV Conductors		
D3	Fittings for LV Conductors		
D4	Colour Marking Table		

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Table 4

• ENA TS 43–15			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
	Type Tests		
E1.2 & 1.3.1	HV Fittings test assemblies		
E1.3.2	Transverse and pull-off tests (Applicable to all Ties)		
E1.3.3	Pull-through withstand tests (Only applicable to Angle Ties)		
E1.3.4	Swinging Conductor tests (Normally only applied to Hazel, Elm & Lynx)		
E1.4	Test Requirements for HV Fittings		

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Table 5

• ENATS 43–15			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
	Type Tests		
E1.7	Test Requirements for LV Fittings		
E1.7.1.1	Failure containment test (Applied to terminations on bare conductors)		
E1.7.1.2	Pull-through withstand test (applied to intermediate fittings on bare conductors)		
E1.7.2.1	Failure containment test (Applied to terminations on PVC covered conductors)		
E1.7.2.2	Pull-through withstand test (applied to intermediate fittings on PVC covered conductors)		
E1.7.3	Minimum hold test (applied to concentric cables)		

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Table 6

• ENATS 43–15			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
E2.2	Inspection Tests - Applicable to all fittings		
E3	Semi-Conducting Tape and Pad Material		
E3.2	Electrochemical test		

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Helical conductor fittings

Table 7

• BS EN 61284			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
4	Requirements (see note 1)		
4.2	Requirements for specific fittings (see note 3)		
5	Quality Assurance		
6	Classification of Tests (see note 5)		
7	Visual Examination		
8	Dimensional and Material Variation		
9	Hot Dip Galvanising		
10	Non destructive Testing		
11	Mechanical Tests		
13	Heat Cycle Tests		
12	Magnetic Loss Test		
	ENA TS 43-92		
4	General requirements (see note 2)		
5	Compression Conductor Fittings (see note 4)		
6	Mechanical Conductor Fittings		
7	Factory formed Helical Conductor Fittings		
8	Quality Assurance (See note 6)		

Note 1 - Clause 4.2 of BS EN 61284 is subject to clauses 5,6 & 7 of ENA TS 43-92.

Note 2 – ENA TS 43-92 calls for specific requirements in relation to:

- finish (ENA TS 43-92 Clause 4.1.3)
- tolerances (ENA TS 43-92 Clause 4.2)
- markings applied to factory formed helical conductor fittings (ENA TS 43-92 Clause 4.3)
- packaging and protection (ENA TS 43-92 clause 4.4)
- tension terminations (ENA TS 43-92 Clause 4.5)

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Notes continued.....

Note 3 - Clause 4.2 of BS EN 61284 covers general design requirements of which shall be complied with, where appropriate to the fittings employed.

Note 4 - Clauses 5,6 & 7 of ENA TS 43-92 call for specific design requirements of compression conductor fitting not called by clause 4.2 of BS EN 61284.

Note 5 - Clause 8.3 of ENA TS 43-92 defines sampling and acceptance criteria.

Note 6 - The appropriate clauses of BS 61284 apply with the following specific requirements:

- Heat cycle test for current carrying fittings shall be 1000 cycles to a temperature rise of 70° above ambient.
- The heat cycle test for current carrying fitting's shall be undertaken on six fittings subject to the requirements of clause 8.1.1 of ENA TS 43-92.
- Clauses 8.2.1, 8.2.2, 8.2.3, 8.2.4 of ENA TS define the conductor tensions to be applied during the mechanical tests and the holding period T.
- Clauses 8.2.2 of ENA TS 43-92 call for additional requirements in terms of the re-application of the fitting to the conductor.
- Clauses 8.2.5 of ENA TS 43-92 calls for additional torque tests for split bolt, saddle fittings, and live line taps.
- Clauses 8.2.7 of ENA TS 43-88 calls for additional tests to prove the interface integrity of friction welded fittings.
- Clause 8.2.9 of ENA TS 43-92 calls method B (vertical damage and failure load tests) for suspension clamps and defines specific application and slippage criteria.

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Tests for Stay Fittings

Table 8

• BS EN 61284			
Clause / Sub-clause	Requirement	Conformance Code	Remarks
7 8 9	Visual Examination Dimensional and Material Variation Hot Dip Galvanising		
ENA TS 43-91			
9	Factory formed Helical Stay Fittings (See note 1)		
8	Quality Assurance		

Note 1 - The appropriate clauses of BS 61284 apply with the following specific requirements:

- Clauses 9.2 of ENA TS 93-91 define the conductor tensions to be applied during the mechanical tests and the holding period T.
- Clauses 9.2 of ENA TS 43-91 call for additional requirements in terms of the re-application of the fitting to the conductor.

Clauses 9.2 of ENATS 43-91 define Sampling and Acceptance Criteria.

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Appendix B – Addendum to Suppliers Requirements

Appendix A contains self certification conformance declarations. They shall be completed by suppliers tendering products for this contract with one set completed for each variant offered.

Fittings shall be supplied in appropriate quantities within cardboard or similar boxes. The fitting and container shall be clearly labelled to allow identification of the product including the CE Electric UK commodity code.

The weight of each package shall not exceed 25kG.